

Visualization Tool for Electric Vehicle Charge and Range Analysis

Abstract

This document describes a visualization tool designed to analyze electric vehicle (EV) battery charge levels and driving range. The tool helps users understand energy consumption, charging behavior, and remaining travel distance through interactive charts and dashboards.

Introduction

Electric vehicles rely on battery capacity, driving conditions, and charging infrastructure. Drivers often face uncertainty about how far they can travel before recharging. This visualization tool provides clear insights into battery usage and range prediction.

Objectives

- Monitor battery charge level in real time
- Estimate remaining driving range
- Analyze energy consumption patterns
- Visualize charging sessions
- Support route planning decisions

System Overview

The tool collects data from the vehicle or simulation inputs, including:

- Battery State of Charge (SoC)
- Energy consumption (kWh/km)
- Vehicle speed and distance
- Charging rate and duration
- Environmental factors (temperature, terrain)

Key Features

1. Real-Time Charge Monitoring

Displays current battery percentage with graphical indicators.

2. Range Prediction

Calculates estimated distance remaining based on driving efficiency.

3. Consumption Analysis

Shows energy usage over time using line or bar charts.

4. Charging Visualization

Tracks charging speed, duration, and energy added during sessions.

5. Historical Data View

Allows comparison of past trips and charging behavior.

Visualization Components

- Gauge charts for battery level
- Line graphs for energy consumption
- Bar charts for trip comparisons
- Maps for route-based range estimation

Benefits

- Reduces range anxiety
- Improves trip planning

- Encourages efficient driving
- Helps optimize charging strategies

Applications

- Electric vehicle dashboards
- Fleet management systems
- Research and development
- Mobile EV monitoring apps

Conclusion

The visualization tool provides a comprehensive view of electric vehicle charge status and range performance. By transforming raw battery data into intuitive visual insights, it supports smarter driving decisions and enhances user confidence in EV technology.